Vishwashankar T. Janakiraman

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Summary

Senior Data Scientist with 5+ years of experience designing scalable data pipelines, building ML and LLM-driven solutions, and applying graph analytics for risk detection. Expertise in Python, PySpark, SQL, and the AWS ecosystem, with strong statistical modeling and A/B testing skills. Proven at translating complex data into actionable insights to drive strategic decision-making.

Technical Skills

Languages: Python, PySpark, R, Bash/sh, Java, SQL
Data Services: Databricks, AWS, Dataleap, Snowflake, Plotly, Tableau.
Database: MongoDB, DynamoDB, PostgreSQL, Hive, ElasticSearch, MySQL, OracleDB, Redis, Neo4j.
Graph Analytics: Community Detection, Node Embedding, Graph Convolutional Neural Netorks(GCNs), Graph Clustering.
AI/ML: LangChain, LangGraph, LlamaIndex, Spark MLlib, PyTorch, TensorFlow, Keras

Experience

TikTok — New York City, NY, USA
Senior Data Scientist/Developer (ByteCloud, PySpark, Docker, Dataleap) — Feb 2024 – Present

Financial Crime Compliance

- Architected and deployed an **AI Agent** based on a **Retrieval-Augmented Generation (RAG) LLM** framework for an AML virtual assistant that reduced report generation time by 90% and enabled near **real-time queries** for risk triage
- Architected the development and deployment of a graph-based **financial crime detection system** using custom algorithms like **Community Detection** and Graph Traversal for risk scoring and entity resolution across **150M+** entities
- Conducted **A/B testing**, **drift analysis**, and **parameter tuning** on financial crime models, establishing robust decision boundaries and managing risk coverage across **500M+** monthly transactions.
- Successfully tuned the production **anomaly detection system** (using Isolation Forest) for the TikTok Live network, leading to a significant **20%** reduction in **false positives** while maintaining detection accuracy.
- Deployed **LLM models** and vector embeddings via ByteCloud containers on NVIDIA H100 GPUs, developing **low-latency Internal API** endpoints **5K QPM** to maintain **security compliance** by avoiding external model hosting
- Automated the discovery and classification of sensitive PII data across **2M+** internal databases by leveraging **ML-based entity recognition** and fine-tuned **LLM classification** models using predefined Data Taxonomy tags, to reduced manual efforts by **80%.**

ADP Inc. — Parsippany, NJ, USA
Data Scientist/Engineer (AWS, Databricks, Snowflake, Airflow) — July 2021 – Feb 2024

- Designed and developed complex **ETL workflows**, **data pipelines** using **Databricks**, **AWS** State Machines, and Glue to streamline data processing from diverse sources into a centralized data lake and performed thorough end-to-end testing.
- Automated **ETL orchestration** using AWS Step Functions with S3 Aurora tables and **DynamoDB**, improving performance and efficiency, achieving a high optimization rate and facilitating quicker data access and processing
- Built real-time text classification models using **AWS SageMaker** for earnings, deductions, and other use cases, integrating automated **ML classification** endpoints and hosted it as API endpoints.
- Enhanced storage efficiency with data partitioning strategies in **S3** and Aurora, and implemented automated **data quality** checks in PySpark-Databricks to proactively **detect anomalies**.
- Collaborated with cross-functional teams to translate business requirements into technical solutions, **reducing costs**.

ADP Inc. — Roseland, NJ, USA
Data Science Intern (PostgreSQL, NetworkX, Plotly Dashboard, AWS) — June 2020 – Aug 2020

- Optimized the **caching strategy** of a data retrieval system by using NLP techniques on unstructured network queries.
- Performed **advanced analytics** using **Python** to obtain network and user features.
- Designed prototypes and obtained additional features using **Graph Clustering** and Community Detection methods for inferring network and user statistics.
- Performed extensive data analysis to develop Community Detection and clustering algorithms for inferring Network Statistics

Infosys Ltd. — Mysore, KA, India
Software Developer Intern (Angular, Spring, OracleDB) — Dec 2018 – May 2019

- Developed a **web application** for the security department to handle employee complaints, which integrated **OracleDB**, **REST APIs**, and Angular, resulting in improved data management efficiency
- Delivered the project within 80% of the proposed duration using **agile** methodology, significantly enhancing complaint resolution efficiency.

Education

New Jersey Institute of Technology — May 2021
Master of Science in Data Science — *GPA: 3.75/4.0*

Amrita Vishwa Vidyapeetham University — May 2019
Bachelors of Technology in Computer Science and Engineering (CSE) — *GPA: 3.8/4.0*